If $x = 4$, $y = \frac{4(4+1)}{2} = \frac{20}{2} = 10$

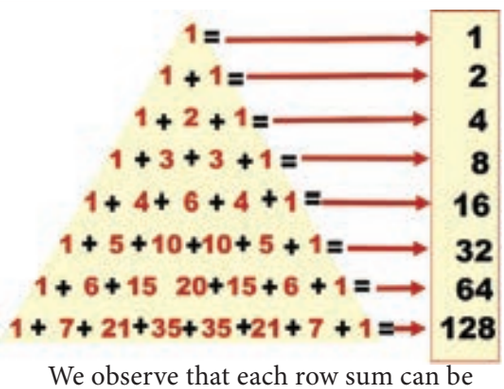
If $x = 5$, $y = \frac{5(5+1)}{2} = \frac{30}{2} = 15$

Hence, $y = \frac{x(x+1)}{2}$ is verified.

Example 5.3 Can row sum of elements in a Pascal's Triangle form a pattern?

Solution

The row sum of elements of a Pascal's Triangle are shown below:



We observe that each row sum can be expressed in powers of 2.

First row $= 2^{1-1} = 1$

Second row $= 2^{2-1} = 2 \times 1 = 2$

Third row $= 2^{3-1} = 2 \times 2 = 4$

Fourth row $= 2^{4-1} = 2 \times 2 \times 2 = 8$

Fifth row $= 2^{5-1} = 2 \times 2 \times 2 \times 2 = 16$

Sixth row $= 2^{6-1} = 2 \times 2 \times 2 \times 2 \times 2 = 32$

Seventh row $= 2^{7-1} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$

Eighth row $= 2^{8-1} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 128$

Here x denotes the row and y denotes the corresponding row sum. The values of x and y can be tabulated as follows:

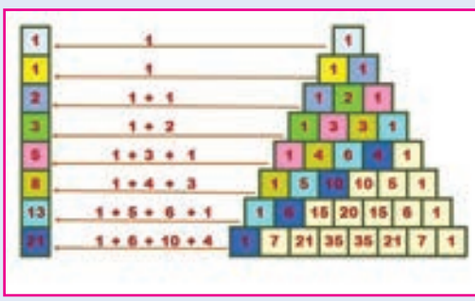
x	1	2	3	4	5	6	7	8	...
y	1	2	4	8	16	32	64	128	...

The relationship between x and y is $y = 2^{x-1}$.



Observe the pattern obtained by adding the elements in the slanting rows of the Pascal's Triangle.

The sequence obtained is known as Fibonacci sequence.



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